

Proximal Balancing

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We study causal adjustment with observed pretreatment covariates X , a high-dimensional pretreatment proxy $W = (W_1, \dots, W_J)$, and an unobserved confounder. For one prespecified block j , the proposed cross-view construction learns a finite representation $Z_j = \phi_j(X, W_{-j})$ without using the held-out block W_j , and measures treatment imbalance through the conditional distribution of (X, W_j) . We prove an exact bias decomposition and show that exact cross-view balance identifies the population average treatment effect under joint latent exchangeability, held-out-view separation, overlap, and outcome-relevant completeness. When balance is approximate, a uniform outcome-relevant stability condition yields the sensitivity certificate $|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j)$. A finite-class learning analysis and an honest augmented inverse probability weighted estimator give an end-to-end finite-sample bound that separates representation error, hidden-confounding sensitivity, nuisance error, and evaluation-sample variation. The causal interpretation depends on the stated cross-view structural conditions; the existence of one generic correlated proxy is not sufficient by itself.

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1. Introduction

1.1. Related Works

Balancing scores and covariate balancing. The propensity score is the canonical reduction of observed pretreatment covariates: under treatment ignorability given the measured covariates, conditioning on the treatment probability balances those covariates across treatment arms and identifies causal effects (Rosenbaum and Rubin, 1983). Entropy balancing and the covariate-balancing propensity score instead choose weights or treatment models that target balance directly (Hainmueller, 2012; Imai and Ratkovic, 2014). These methods motivate our optimization target, but their usual causal guarantee still requires the measured covariates to suffice for confounding control. PROBE instead allows a latent confounder and asks what a held-out proxy block can reveal about the bias remaining after reducing the other pretreatment measurements.

Causal representation learning. Representation-learning methods compress high-dimensional covariates while penalizing imbalance between treated and control representations. For example, counterfactual regression bounds counterfactual prediction error using factual prediction error and an integral probability metric between learned treatment-arm representations (Shalit et al., 2017). PROBE shares the use of a learned low-dimensional representation and a distributional balance penalty. Its distinguishing operation is blockwise exclusion: one proxy block is held out, the representation is learned from the remaining pretreatment information, and conditional treatment balance is evaluated with the held-out block.

Negative controls and proximal causal inference. Negative-control variables can reveal residual confounding because they should not be causally affected by treatment or should not causally affect the outcome, while remaining associated with hidden common causes (Lipsitch et al., 2010). Proximal causal inference turns this diagnostic idea into point identification by assigning distinct treatment-inducing and outcome-inducing proxy roles and imposing completeness and bridge conditions (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Semiparametric proximal theory then develops influence functions and efficient estimators for the resulting identifying functionals (Cui et al., 2024). PROBE takes a different operational route: the remaining proxy blocks form the input to an adjustment representation, while the held-out block measures residual treatment imbalance. An outcome-relevant completeness condition transfers exact observed balance to the potential-outcome means needed for the population average treatment effect.

Single-proxy identification. Single Proxy Control shows that one negative-control outcome can identify the effect of treatment on the treated under a counterfactual proxy condition and suitable bridge structure (Park et al., 2024). This differs from our population average treatment effect target, our latent-confounder proxy interpretation, and our use of an observable balance discrepancy. The comparison makes clear that “one proxy” does not by itself identify a common estimand. Identification depends on what the proxy measures, which conditional independence is imposed, and whether the analysis uses a bridge or an adjustment representation checked by a held-out proxy block.

Kernel discrepancies and sensitivity certification. Maximum mean discrepancy is an integral probability metric that equals zero only for identical distributions when its kernel is characteristic (Gretton et al., 2012). PROBE uses conditional treatment-arm discrepancies of the measured covariates and the held-out block as observable magnitudes, not kernel-test p -values. Multiplying this discrepancy by an outcome-relevant stability constant yields a bias certificate and a representation-

selection guarantee. The individual ingredients have close precedents in balancing, representation learning, kernel testing, and proximal inference. We therefore state the contribution through the precise assumptions, estimand, and guarantees below rather than making an unsupported priority claim.

2. Problem Setting

2.1. Preliminaries

Structural causal models. A structural causal model (SCM) consists of endogenous variables, exogenous disturbances, and structural assignments. Writing the endogenous variables as $S_1, \dots, S_d$, an acyclic SCM has equations

$$S_k = f_k(\text{pa}_k, E_k), \quad k = 1, \dots, d, \quad (1)$$

where pa_k contains the direct causes of S_k and E_k is an exogenous disturbance. The intervention $\text{do}(A = a)$ replaces the structural equation for A by the constant a while leaving the other equations unchanged. The resulting outcome is the potential outcome $Y(a)$. An average causal effect is *identified* when its value is uniquely determined by the observed-data distribution for every SCM in the stated model class (Pearl, 2009).

Proximal causal inference. Ordinary adjustment seeks observed covariates that block all non-causal paths from treatment to outcome. Proximal causal inference instead allows an unobserved confounder and uses proxy measurements of it. Standard proximal formulations assign two different proxy roles and posit a bridge equation whose solution translates an observable conditional moment into the target causal functional (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Their completeness condition requires the relevant conditional-expectation operator to be injective, so that no nonzero latent function is invisible to all proxy conditional moments. This condition does not follow from the mere existence of a correlated proxy. PROBE instead uses outcome-relevant completeness to interpret exact balance measured through a held-out proxy block and a quantitative stability condition when balance is only approximate.

2.2. Problem Setting

We observe independent copies

$$O_i = (X_i, W_i, A_i, Y_i), \quad i = 1, \dots, n, \quad (2)$$

where X is a possibly high-dimensional vector of observed pretreatment covariates, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. The observed pretreatment measurement W is divided into prespecified blocks,

$$W = (W_1, \dots, W_J). \quad (3)$$

Each block may be scalar or vector-valued. A block can represent one repeated measurement, sensor channel, laboratory panel, or prespecified region of a larger measurement. We call W a candidate

proxy because some or all blocks may contain information about an unobserved pretreatment variable U that affects treatment selection and the outcome.

Fix one block index $j \in \{1, \dots, J\}$ before analyzing the data. We call W_j the *held-out proxy block* and write

$$W_{-j} \triangleq (W_1, \dots, W_{j-1}, W_{j+1}, \dots, W_J) \quad (4)$$

for all remaining proxy blocks. The measured input available for constructing an adjustment representation is

$$V_j \triangleq (X, W_{-j}). \quad (5)$$

The representation itself is introduced only in Section 3.

Figure 1 contrasts the conventional two-proxy roles with a broad graph compatible with the PROBE problem setting.

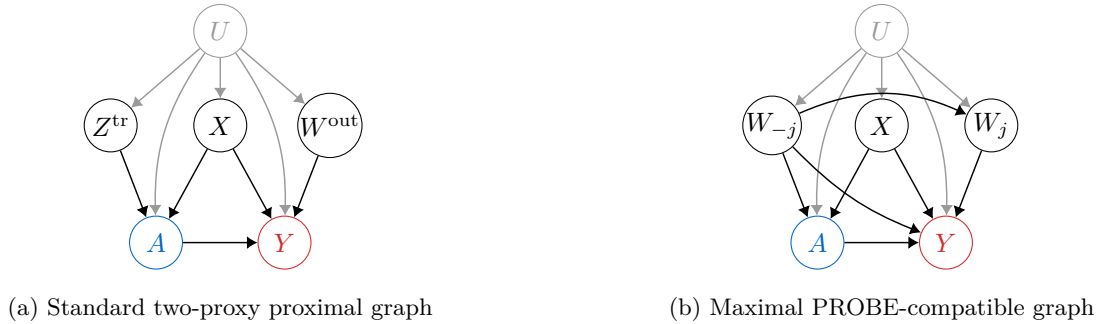


Figure 1: Illustrative maximal causal graphs. Panel (a) displays the conventional treatment-inducing proxy Z^{tr} and outcome-inducing proxy W^{out} . Panel (b) allows the remaining blocks W_{-j} to affect treatment, outcome, and the held-out block, while W_j may affect the outcome but has no direct arrow into treatment. These graphs illustrate permitted causal relations; neither graph is equivalent to the completeness and conditional-independence assumptions stated later.

For each $a \in \{0, 1\}$, let $Y(a)$ denote the outcome that would be observed if treatment were set to a . The target is the population average treatment effect

$$\tau \triangleq \mathbb{E}\{Y(1) - Y(0)\}. \quad (6)$$

We allow U to remain associated with treatment after conditioning on all observed pretreatment measurements. The research question is whether the prespecified block structure can support a learned adjustment representation that identifies τ under exact PROBE balance and controls residual hidden-confounding bias when balance is approximate. The representation and the assumptions that give its balance a causal meaning are introduced next.

3. Identification

We now construct the adjustment representation. Let $\mathcal{X}$, $\mathcal{W}_j$, and $\mathcal{V}_j$ denote the value spaces of X , W_j , and V_j , respectively. Fix a positive integer M_j , write $[M_j] \triangleq \{1, \dots, M_j\}$, and prespecify a class Φ_j of measurable maps from $\mathcal{V}_j \times [0, 1]$ to $[M_j]$. Let

$$\begin{aligned} \epsilon_j &\sim \text{Unif}(0, 1), \\ \epsilon_j &\perp\!\!\!\perp \{U, V_j, W_j, A, Y(0), Y(1)\} \end{aligned} \quad (7)$$

and, for a fixed $\phi_j \in \Phi_j$, define

$$Z_j \triangleq \phi_j(V_j, \epsilon_j). \quad (8)$$

The auxiliary variable ϵ_j permits a stochastic representation. A deterministic representation is the special case in which ϕ_j ignores its second argument. The finite range makes balance, overlap, and adjustment explicit within strata, but does not require the hidden variable U to be discrete. A value $z \in [M_j]$ is retained when $P(Z_j = z) > 0$.

3.1. Assumptions

Assumption 1 (Consistency and integrability). *For each $a \in \{0, 1\}$,*

$$Y = Y(A), \quad \mathbb{E}|Y(a)| < \infty. \quad (9)$$

Assumption 2 (Latent exchangeability and a common held-out channel). *For each $a \in \{0, 1\}$, the following two clauses hold separately:*

$$Y(a) \perp\!\!\!\perp A \mid (U, V_j), \quad (10)$$

$$W_j \perp\!\!\!\perp A \mid (U, V_j). \quad (11)$$

The first clause says that the hidden state U and the full representation input $V_j = (X, W_{-j})$ suffice to remove confounding of treatment and each potential outcome. The second says that, after the same variables are fixed, the conditional measurement law of the held-out block is common to both treatment arms. These clauses allow W_{-j} to affect treatment and outcome. They do not assert the stronger joint independence $A \perp\!\!\!\perp \{Y(a), W_j\} \mid (U, V_j)$, which is not needed below.

Assumption 3 (Representation overlap). *For every retained representation value z ,*

$$0 < P(A = 1 \mid Z_j = z) < 1. \quad (12)$$

Both treatment arms therefore occur with positive probability in every retained stratum.

We next define the objects required to state the last identification assumption. For $a \in \{0, 1\}$ and

any retained z , let

$$Q_{a,j,z} \triangleq \mathcal{L}(V_j, U \mid A = a, Z_j = z). \quad (13)$$

This law records both the remaining variation in the representation input and the hidden state within one treatment arm and one representation stratum.

For every measurable $C \subseteq \mathcal{W}_j$, define the held-out conditional law

$$K_j(C \mid v, u) \triangleq P(W_j \in C \mid V_j = v, U = u). \quad (14)$$

For a probability distribution Q on (V_j, U) and a measurable $B \subseteq \mathcal{X} \times \mathcal{W}_j$, define

$$(\mathcal{K}_j Q)(B) \triangleq \int \left[\int 1\{(x, w_j) \in B\} K_j\{dw_j \mid (x, w_{-j}), u\} \right] Q(dx, dw_{-j}, du). \quad (15)$$

Thus, $\mathcal{K}_j Q$ is the induced distribution of (X, W_j) : draw $(V_j, U) = (X, W_{-j}, U)$ from Q , generate W_j through K_j , and retain only (X, W_j) .

For each treatment level and retained stratum, define

$$\mu_{a,j,z}(v, u) \triangleq \mathbb{E}\{Y(a) \mid V_j = v, U = u, Z_j = z\}. \quad (16)$$

Let

$$S_{j,z} \triangleq \text{supp}\{\mathcal{L}(V_j, U \mid Z_j = z)\} \quad (17)$$

and let $\mathcal{Q}_{j,z}$ contain all probability distributions Q supported on $S_{j,z}$ for which the two means below are finite. For $Q \in \mathcal{Q}_{j,z}$, write

$$\mathbb{E}_Q\{\mu_{a,j,z}(V_j, U)\} \triangleq \int \mu_{a,j,z}(v, u) Q(dv, du). \quad (18)$$

The realized arm laws $Q_{0,j,z}$ and $Q_{1,j,z}$ are required to belong to $\mathcal{Q}_{j,z}$.

Assumption 4 (Exact outcome-relevant completeness). *For every retained z , every $Q, Q' \in \mathcal{Q}_{j,z}$, and every $a \in \{0, 1\}$,*

$$\mathcal{K}_j Q = \mathcal{K}_j Q' \implies \mathbb{E}_Q\{\mu_{a,j,z}(V_j, U)\} = \mathbb{E}_{Q'}\{\mu_{a,j,z}(V_j, U)\}. \quad (19)$$

Two admissible laws that are indistinguishable through the induced distribution of (X, W_j) must agree on each outcome-relevant mean. This condition is imposed over the entire class $\mathcal{Q}_{j,z}$ defined above, not only on the two realized arm laws. It does not require $Q = Q'$ and therefore does not imply latent balance. On the same admissible class it is weaker than full mixture injectivity, which would require $\mathcal{K}_j Q = \mathcal{K}_j Q'$ to imply $Q = Q'$. It is not uniformly comparable to standard proximal completeness because the two conditions act on different operators and domains.

3.2. Exact Cross-View Identification

Lemma 1 (Conditioning safety). *Under Assumption 2, for each $a \in \{0, 1\}$,*

$$\mathbb{E}\{Y(a) \mid A, V_j, U, Z_j\} = \mathbb{E}\{Y(a) \mid V_j, U, Z_j\}. \quad (20)$$

Under the common-channel clause of the same assumption, for every measurable C ,

$$P(W_j \in C \mid A, V_j, U, Z_j) = K_j(C \mid V_j, U). \quad (21)$$

Proof. Conditional on V_j , the representation Z_j depends only on the auxiliary random variable ϵ_j . By the explicit independence condition imposed on ϵ_j above, adding Z_j to the conditioning set does not introduce information about A , $Y(a)$, or W_j beyond (V_j, U) . Applying the two separate clauses of Assumption 2 therefore gives the two displayed conditional-law equalities. Equation (14) identifies the common conditional law in the second equality. $\square$

Definition 1 (Adjustment functional). *For the fixed representation Z_j , define*

$$\tau_{Z_j} \triangleq \sum_{z: P(Z_j=z)>0} P(Z_j = z) [\mathbb{E}(Y \mid A = 1, Z_j = z) - \mathbb{E}(Y \mid A = 0, Z_j = z)]. \quad (22)$$

This is an observed-data functional. Its equality with τ has not yet been asserted.

For each retained stratum, write

$$p_{j,z} \triangleq P(Z_j = z), \quad \pi_{j,z} \triangleq P(A = 1 \mid Z_j = z). \quad (23)$$

Theorem 1 (Exact bias decomposition). *Under Assumptions 1–3,*

$$\begin{aligned} \tau_{Z_j} - \tau = \sum_z p_{j,z} & \left[(1 - \pi_{j,z}) \left[\mathbb{E}_{Q_{1,j,z}} \{\mu_{1,j,z}\} - \mathbb{E}_{Q_{0,j,z}} \{\mu_{1,j,z}\} \right] \right. \\ & \left. + \pi_{j,z} \left[\mathbb{E}_{Q_{1,j,z}} \{\mu_{0,j,z}\} - \mathbb{E}_{Q_{0,j,z}} \{\mu_{0,j,z}\} \right] \right]. \end{aligned} \quad (24)$$

Proof. Consistency and Lemma 1 give

$$\mathbb{E}(Y \mid A = a, Z_j = z) = \mathbb{E}_{Q_{a,j,z}} \{\mu_{a,j,z}(V_j, U)\}. \quad (25)$$

The law of total expectation over treatment gives

$$\begin{aligned} \mathbb{E}\{Y(a) \mid Z_j = z\} &= \pi_{j,z} \mathbb{E}_{Q_{1,j,z}} \{\mu_{a,j,z}(V_j, U)\} \\ &+ (1 - \pi_{j,z}) \mathbb{E}_{Q_{0,j,z}} \{\mu_{a,j,z}(V_j, U)\}. \end{aligned} \quad (26)$$

Substitute these identities into τ_{Z_j} and τ , subtract, and collect the coefficients of the two arm laws. This yields equation (24). $\square$

Lemma 2 (Cross-view factorization). *Under the common-channel clause of Assumption 2,*

$$\mathcal{L}(X, W_j \mid A = a, Z_j = z) = \mathcal{K}_j Q_{a,j,z} \quad (27)$$

for every treatment arm and retained stratum.

Proof. For any measurable $B \subseteq \mathcal{X} \times \mathcal{W}_j$, iterated expectation over (V_j, U) gives

$$\begin{aligned} & P\{(X, W_j) \in B \mid A = a, Z_j = z\} \\ &= \int \left[\int 1\{(x, w_j) \in B\} P\{dw_j \mid A = a, V_j = (x, w_{-j}), U = u, Z_j = z\} \right] Q_{a,j,z}(dx, dw_{-j}, du). \end{aligned} \quad (28)$$

Lemma 1 replaces the inner conditional law by $K_j\{dw_j \mid (x, w_{-j}), u\}$. Equation (15) then gives the stated factorization. $\square$

Definition 2 (Exact PROBE balance). *The fixed representation satisfies exact PROBE balance when*

$$(X, W_j) \perp\!\!\!\perp A \mid Z_j. \quad (29)$$

Under representation overlap, this condition is equivalent to equality of the two conditional laws of (X, W_j) in every retained stratum.

Remark 1 (Why copying the representation input is not an automatic solution). *Outside the finite-range representation class, consider the non-compressive benchmark $Z_j = V_j$. Then balance of X is automatic because X is contained in V_j . Balance of the held-out block is not automatic. Exact PROBE balance reduces to*

$$W_j \perp\!\!\!\perp A \mid V_j. \quad (30)$$

Thus, self-copying is permitted by the identification theory but does not automatically pass the observable balance endpoint. Representation learning instead seeks a lower-dimensional tradeoff between held-out balance and representation overlap.

Exact PROBE balance and Lemma 2 imply

$$\mathcal{K}_j Q_{1,j,z} = \mathcal{K}_j Q_{0,j,z} \quad (31)$$

for every retained stratum.

Theorem 2 (Exact identification under PROBE balance). *Suppose Assumptions 1–4 hold for the fixed block and representation. If Z_j satisfies exact PROBE balance, then, for each $a \in \{0, 1\}$ and retained z ,*

$$\mathbb{E}\{Y(a) \mid A = 1, Z_j = z\} = \mathbb{E}\{Y(a) \mid A = 0, Z_j = z\}. \quad (32)$$

Consequently,

$$\tau_{Z_j} = \tau, \quad (33)$$

and the population ATE is point identified by equation (22).

Proof. Fix a retained stratum. Exact PROBE balance and Lemma 2 give equation (31). Both arm laws belong to $\mathcal{Q}_{j,z}$, so outcome-relevant completeness gives

$$\mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(V_j, U)\} = \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(V_j, U)\} \quad (34)$$

for each treatment level a . Lemma 1 identifies these two quantities as the corresponding arm-specific conditional means of $Y(a)$, proving equation (32). Both differences in the exact bias decomposition are therefore zero, which proves $\tau_{Z_j} = \tau$. $\square$

Remark 2 (What is, and is not, identified). *The theorem establishes conditional mean exchangeability, which is sufficient for ATE adjustment. It does not assert the stronger conditional independence $Y(a) \perp\!\!\!\perp A \mid Z_j$, equality of the arm laws $Q_{1,j,z} = Q_{0,j,z}$, or latent balance.*

Remark 3 (Attainability). *The theorem is conditional on a representation attaining exact PROBE balance. It does not guarantee that such a representation exists in Φ_j . A restricted class may leave positive residual imbalance, which motivates the next subsection.*

3.3. Approximate Cross-View Bias Bound

Let κ_j be a bounded, measurable, positive-semidefinite kernel on the support of (X, W_j) , and let $\mathcal{H}_j$ be its reproducing kernel Hilbert space. For a distribution P of (X, W_j) , define its kernel mean embedding

$$m_j(P) \triangleq \int \kappa_j(r, \cdot) P(dr) \in \mathcal{H}_j. \quad (35)$$

For two such distributions P and P' , define

$$\text{MMD}_{\kappa_j}(P, P') \triangleq \|m_j(P) - m_j(P')\|_{\mathcal{H}_j}. \quad (36)$$

Assume that κ_j is characteristic, meaning that this discrepancy is zero exactly when $P = P'$.

Definition 3 (Cross-view discrepancy). *For each retained stratum, define*

$$\Delta_{j,z}(\phi_j) \triangleq \text{MMD}_{\kappa_j} \{ \mathcal{L}(X, W_j \mid A = 1, Z_j = z), \mathcal{L}(X, W_j \mid A = 0, Z_j = z) \}. \quad (37)$$

The aggregate discrepancy is

$$D_j(\phi_j) \triangleq \sum_z p_{j,z} \Delta_{j,z}(\phi_j), \quad (38)$$

where the sum is over retained strata.

Proposition 1 (Zero discrepancy and exact balance). *Under representation overlap and the characteristic-kernel condition,*

$$D_j(\phi_j) = 0 \iff (X, W_j) \perp\!\!\!\perp A \mid Z_j. \quad (39)$$

Proof. Every summand is nonnegative and every retained stratum has positive mass. Thus, $D_j(\phi_j) = 0$ exactly when every stratum-specific MMD is zero. Characteristicity converts each zero MMD into equality of the two conditional laws, and overlap makes those equalities equivalent to conditional independence. $\square$

The factorization lemma also gives

$$\Delta_{j,z}(\phi_j) = \text{MMD}_{\kappa_j}(\mathcal{K}_j Q_{1,j,z}, \mathcal{K}_j Q_{0,j,z}). \quad (40)$$

Assumption 5 (Uniform outcome-relevant stability). *There is a finite constant Γ_j such that, for every retained z , every $a \in \{0, 1\}$, and every $Q, Q' \in \mathcal{Q}_{j,z}$,*

$$\begin{aligned} & |\mathbb{E}_Q\{\mu_{a,j,z}(V_j, U)\} - \mathbb{E}_{Q'}\{\mu_{a,j,z}(V_j, U)\}| \\ & \leq \Gamma_j \text{MMD}_{\kappa_j}(\mathcal{K}_j Q, \mathcal{K}_j Q'). \end{aligned} \quad (41)$$

The constant Γ_j converts an observed discrepancy into a bound on an outcome-relevant latent difference. It is a structural sensitivity parameter and is not generally identified from the observed distribution.

Theorem 3 (Approximate cross-view bias bound). *Suppose Assumptions 1–3 and Assumption 5 hold for the fixed representation. Then*

$$|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j). \quad (42)$$

Proof. For each treatment level and retained stratum, define

$$\delta_{a,j,z} \triangleq \mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(V_j, U)\} - \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \quad (43)$$

The exact bias decomposition and the triangle inequality give

$$|\tau_{Z_j} - \tau| \leq \sum_z p_{j,z} [(1 - \pi_{j,z})|\delta_{1,j,z}| + \pi_{j,z}|\delta_{0,j,z}|]. \quad (44)$$

Uniform stability and equation (40) imply

$$|\delta_{a,j,z}| \leq \Gamma_j \Delta_{j,z}(\phi_j). \quad (45)$$

Substitution and $(1 - \pi_{j,z}) + \pi_{j,z} = 1$ yield equation (42). $\square$

Remark 4 (Exact completeness and quantitative stability). *Outcome-relevant completeness preserves equality at zero discrepancy. Uniform stability controls nonzero discrepancies and therefore implies outcome-relevant completeness. The converse need not hold because equality at zero can be preserved even when no finite Lipschitz constant Γ_j exists.*

Remark 5 (Sensitivity interpretation). *For a scientifically specified Γ_j , the approximate bound gives*

$$\tau \in [\tau_{Z_j} - \Gamma_j D_j(\phi_j), \tau_{Z_j} + \Gamma_j D_j(\phi_j)]. \quad (46)$$

A small discrepancy alone does not guarantee small causal bias. The causal quantity is the product $\Gamma_j D_j(\phi_j)$.

Remark 6 (Uniformity across candidate representations). *For representation learning, let $\Gamma_j(\phi)$ be a stability budget for candidate ϕ . Comparing candidates by discrepancy requires*

$$\sup_{\phi \in \Phi_j} \Gamma_j(\phi) \leq \bar{\Gamma}_j < \infty. \quad (47)$$

Without this uniform condition, a representation can have a small observed discrepancy and an arbitrarily large stability constant.

4. Representation Learning and AIPW Estimation

The identification results concern one fixed representation. We now select a representation from data and estimate the adjustment functional that it induces. The analysis uses one prespecified held-out block j throughout. For finite-sample analysis, this section restricts equation (8) to the deterministic subclass. We identify each deterministic map with $v \mapsto \phi(v, 0)$ and write it simply as $\phi(v)$.

4.1. Representation Learning

Partition the observed sample into three mutually disjoint subsets,

$$\mathcal{I}_D, \quad \mathcal{I}_N, \quad \mathcal{I}_E, \quad (48)$$

with respective sizes n_D , n_N , and n_E . The discrepancy-learning sample $\mathcal{I}_D$ selects a representation. The nuisance-learning sample $\mathcal{I}_N$ estimates the outcome regressions and propensity used by AIPW.

The evaluation sample $\mathcal{I}_E$ calculates the final estimate. Conditional on the preceding samples, the objects evaluated on the next sample are fixed.

For the finite-sample learning result, fix a finite candidate library

$$\mathcal{F}_j \subseteq \Phi_j, \quad N_j \triangleq |\mathcal{F}_j| < \infty. \quad (49)$$

For $\phi \in \mathcal{F}_j$, define

$$Z_j^\phi \triangleq \phi(X, W_{-j}) \quad (50)$$

and let $\mathcal{R}_j(\phi)$ be its set of positive-mass strata. Assume there are constants $p_{\min} > 0$ and $\eta \in (0, 1/2]$ such that every candidate and every $z \in \mathcal{R}_j(\phi)$ satisfy

$$P(Z_j^\phi = z) \geq p_{\min}, \quad (51)$$

$$\eta \leq P(A = 1 \mid Z_j^\phi = z) \leq 1 - \eta. \quad (52)$$

Assume also that the bounded kernel satisfies

$$\sup_v \sqrt{\kappa_j(v, v)} \leq \rho_j < \infty. \quad (53)$$

The primitive cross-view Assumptions 2 and 2 do not depend on the selected candidate. Representation overlap and outcome-relevant stability must hold for every candidate-specific representation. Their stability budgets are required to obey the common bound in equation (47).

Definition 4 (Empirical cross-view discrepancy). *For observation $i \in \mathcal{I}_D$ and candidate $\phi \in \mathcal{F}_j$, write*

$$Z_{j,i}^\phi \triangleq \phi(X_i, W_{-j,i}). \quad (54)$$

For treatment arm a and stratum z , define

$$N_{a,z}^\phi \triangleq \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\}, \quad (55)$$

$$N_z^\phi \triangleq N_{0,z}^\phi + N_{1,z}^\phi, \quad (56)$$

$$\hat{p}_{j,z}(\phi) \triangleq \frac{N_z^\phi}{n_D}. \quad (57)$$

When $N_{a,z}^\phi > 0$, define the empirical kernel mean

$$\hat{m}_{a,j,z}(\phi) \triangleq \frac{1}{N_{a,z}^\phi} \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\} \kappa_j\{(X_i, W_{j,i}), \cdot\}. \quad (58)$$

When both treatment-arm cells are nonempty, define

$$\hat{\Delta}_{j,z}(\phi) \triangleq \|\hat{m}_{1,j,z}(\phi) - \hat{m}_{0,j,z}(\phi)\|_{\mathcal{H}_j} \quad (59)$$

and

$$\widehat{D}_j(\phi) \triangleq \sum_{z: N_z^\phi > 0} \widehat{p}_{j,z}(\phi) \widehat{\Delta}_{j,z}(\phi). \quad (60)$$

If an occupied empirical stratum contains only one treatment arm, set $\widehat{D}_j(\phi) = +\infty$.

Proposition 2 (Uniform deviation over the finite library). *Let $\delta_D \in (0, 1)$ and suppose*

$$n_D \geq \frac{8}{p_{\min} \eta} \log \left(\frac{8M_j N_j}{\delta_D} \right). \quad (61)$$

Define

$$e_{j,n_D}(\delta_D) \triangleq \rho_j \left[2 + \sqrt{2 \log \left(\frac{4M_j N_j}{\delta_D} \right)} \right] \sqrt{\frac{2}{n_D p_{\min} \eta}} \quad (62)$$

and

$$\begin{aligned} r_{j,n_D}(\delta_D) &\triangleq 2e_{j,n_D}(\delta_D) \\ &\quad + 2\rho_j \sqrt{\frac{2 \left[M_j \log 2 + \log \left(\frac{4N_j}{\delta_D} \right) \right]}{n_D}}. \end{aligned} \quad (63)$$

Under equations (51) through (53), with probability at least $1 - \delta_D$,

$$\boxed{\sup_{\phi \in \mathcal{F}_j} |\widehat{D}_j(\phi) - D_j(\phi)| \leq r_{j,n_D}(\delta_D)}. \quad (64)$$

Proof. Equation (61) and a multiplicative Chernoff bound imply that every candidate treatment-by-stratum cell contains at least $n_D p_{\min} \eta / 2$ observations except on an event of probability at most $\delta_D / 4$. Conditional on the cell assignments, bounded Hilbert-space mean concentration and a union bound over at most $2M_j N_j$ cells give

$$|\widehat{\Delta}_{j,z}(\phi) - \Delta_{j,z}(\phi)| \leq 2e_{j,n_D}(\delta_D) \quad (65)$$

simultaneously, with failure probability at most $\delta_D / 2$. A multinomial L_1 concentration bound gives

$$\sum_z |\widehat{p}_{j,z}(\phi) - p_{j,z}(\phi)| \leq \sqrt{\frac{2 \left[M_j \log 2 + \log (4N_j / \delta_D) \right]}{n_D}} \quad (66)$$

simultaneously over candidates, with failure probability at most $\delta_D / 4$. On the intersection of the three events, use $0 \leq \Delta_{j,z}(\phi) \leq 2\rho_j$ and the triangle inequality to obtain equation (64). The three failure probabilities sum to δ_D . $\square$

Definition 5 (Approximate empirical minimizer). *Let $\varepsilon_{\text{opt},j} \geq 0$. The representation learner returns $\hat{\phi}_j \in \mathcal{F}_j$ satisfying*

$$\hat{D}_j(\hat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} \hat{D}_j(\phi) + \varepsilon_{\text{opt},j}. \quad (67)$$

The learned representation is

$$\hat{Z}_j \triangleq \hat{\phi}_j(X, W_{-j}). \quad (68)$$

The tolerance records optimization error and is zero for an exact empirical minimizer.

Theorem 4 (Representation-learning oracle inequality). *Suppose the conditions of Proposition 2 hold. Suppose Assumptions 1 through 2 hold for block j , and suppose representation overlap and uniform outcome-relevant stability hold for every candidate with the common budget $\bar{\Gamma}_j$. Then, with probability at least $1 - \delta_D$,*

$$D_j(\hat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j}. \quad (69)$$

On the same event,

$$|\tau_{\hat{Z}_j} - \tau| \leq \bar{\Gamma}_j \left[\inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j} \right]. \quad (70)$$

Proof. On the uniform-deviation event,

$$D_j(\hat{\phi}_j) \leq \hat{D}_j(\hat{\phi}_j) + r_{j,n_D}(\delta_D) \quad (71)$$

$$\leq \inf_{\phi \in \mathcal{F}_j} \hat{D}_j(\phi) + \varepsilon_{\text{opt},j} + r_{j,n_D}(\delta_D) \quad (72)$$

$$\leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j}. \quad (73)$$

This proves equation (69). Conditional on $\mathcal{I}_D$, the selected map is fixed. Theorem 3 and the common stability budget give $|\tau_{\hat{Z}_j} - \tau| \leq \bar{\Gamma}_j D_j(\hat{\phi}_j)$. Substitution proves equation (70). $\square$

Remark 7 (Practical learner outside the finite-library theorem). *The theorem covers hard, finite-valued representations in a finite prespecified library. A neural learner using random features, soft assignments, and Adam requires additional bounds for neural-class complexity, feature approximation, soft-to-hard transfer, and optimization error. These guarantees are not supplied by Theorem 4.*

4.2. AIPW Estimation

Conditional on $\mathcal{I}_D$, the map $\hat{\phi}_j$ is fixed. For a fresh observation, define the conditional outcome means and representation propensity

$$m_{a,j}(z) \triangleq \mathbb{E}(Y \mid A = a, \hat{Z}_j = z), \quad (74)$$

$$e_j(z) \triangleq P(A = 1 \mid \hat{Z}_j = z). \quad (75)$$

For the observed adjustment functional $\tau_{\hat{Z}_j}$, the efficient influence function in the nonparametric model for $(\hat{Z}_j, A, Y)$ is

$$\begin{aligned} \psi_j(O) &\triangleq m_{1,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j) \\ &\quad + \frac{A\{Y - m_{1,j}(\hat{Z}_j)\}}{e_j(\hat{Z}_j)} - \frac{(1-A)\{Y - m_{0,j}(\hat{Z}_j)\}}{1 - e_j(\hat{Z}_j)} - \tau_{\hat{Z}_j}. \end{aligned} \quad (76)$$

Use $\mathcal{I}_N$ to estimate the nuisance functions. Denote the estimates by $\hat{m}_{0,j}$, $\hat{m}_{1,j}$, and $\hat{e}_j$.

Definition 6 (Honest AIPW estimator). *For $i \in \mathcal{I}_E$, define $\hat{Z}_{j,i} = \hat{\phi}_j(X_i, W_{-j,i})$ and*

$$\begin{aligned} S_j(O_i) &\triangleq \hat{m}_{1,j}(\hat{Z}_{j,i}) - \hat{m}_{0,j}(\hat{Z}_{j,i}) \\ &\quad + \frac{A_i\{Y_i - \hat{m}_{1,j}(\hat{Z}_{j,i})\}}{\hat{e}_j(\hat{Z}_{j,i})} \\ &\quad - \frac{(1 - A_i)\{Y_i - \hat{m}_{0,j}(\hat{Z}_{j,i})\}}{1 - \hat{e}_j(\hat{Z}_{j,i})}. \end{aligned} \quad (77)$$

The honest AIPW estimator is

$$\hat{\tau}_j^{\text{AIPW}} \triangleq \frac{1}{n_E} \sum_{i \in \mathcal{I}_E} S_j(O_i). \quad (78)$$

Proposition 3 (Exact signed nuisance remainder). *Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$,*

$$\begin{aligned} &\mathbb{E}\{S_j(O)\} - \tau_{\hat{Z}_j} \\ &= \mathbb{E}\left[\{\hat{e}_j(\hat{Z}_j) - e_j(\hat{Z}_j)\} \left\{ \frac{\hat{m}_{1,j}(\hat{Z}_j) - m_{1,j}(\hat{Z}_j)}{\hat{e}_j(\hat{Z}_j)} \right. \right. \\ &\quad \left. \left. + \frac{\hat{m}_{0,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j)}{1 - \hat{e}_j(\hat{Z}_j)} \right\}\right]. \end{aligned} \quad (79)$$

Proof. Fix $\hat{Z}_j = z$. Taking the conditional expectation of the score and using equation (75) gives

$$\begin{aligned} \mathbb{E}\{S_j(O) \mid \hat{Z}_j = z\} &= \hat{m}_{1,j}(z) - \hat{m}_{0,j}(z) \\ &\quad + \frac{e_j(z)\{m_{1,j}(z) - \hat{m}_{1,j}(z)\}}{\hat{e}_j(z)} \end{aligned}$$

$$- \frac{\{1 - e_j(z)\}\{m_{0,j}(z) - \hat{m}_{0,j}(z)\}}{1 - \hat{e}_j(z)}. \quad (80)$$

Subtract $m_{1,j}(z) - m_{0,j}(z)$ and collect the two nuisance errors. Averaging the resulting identity over $\hat{Z}_j$ proves equation (79). $\square$

The remainder is zero if the propensity estimator is correct. It is also zero if both conditional outcome estimators are correct.

Theorem 5 (Conditional finite-sample AIPW error). *Suppose $|Y| \leq B$ almost surely. Clip the nuisance estimates so that*

$$-B \leq \hat{m}_{a,j}(z) \leq B, \quad \eta \leq \hat{e}_j(z) \leq 1 - \eta. \quad (81)$$

For a function g of $\hat{Z}_j$, define

$$\|g\|_{2,\hat{Z}_j} \triangleq \left[\mathbb{E}\{g(\hat{Z}_j)^2 \mid \mathcal{I}_D, \mathcal{I}_N\} \right]^{1/2}. \quad (82)$$

After the two learning samples are observed, define

$$q_{e,j} \triangleq \|\hat{e}_j - e_j\|_{2,\hat{Z}_j}, \quad (83)$$

$$q_{a,j} \triangleq \|\hat{m}_{a,j} - m_{a,j}\|_{2,\hat{Z}_j}, \quad a \in \{0, 1\}. \quad (84)$$

For $\delta_E \in (0, 1)$, define

$$\begin{aligned} \varepsilon_{\text{AIPW},j} &\triangleq 2B \left(1 + \frac{1}{\eta} \right) \sqrt{\frac{2 \log(2/\delta_E)}{n_E}} \\ &\quad + \frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \end{aligned} \quad (85)$$

Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$, with probability at least $1 - \delta_E$ over $\mathcal{I}_E$,

$$\boxed{|\hat{\tau}_j^{\text{AIPW}} - \tau_{\hat{Z}_j}| \leq \varepsilon_{\text{AIPW},j}}. \quad (86)$$

Proof. The boundedness and clipping conditions imply

$$|S_j(O)| \leq 2B \left(1 + \frac{1}{\eta} \right). \quad (87)$$

Conditional on the two learning samples, Hoeffding's inequality bounds the difference between the evaluation average and its conditional expectation by the first term in equation (85). Proposition 3, Cauchy-Schwarz, and the propensity clipping bound control the absolute conditional remainder by

$$\frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \quad (88)$$

The triangle inequality proves equation (86). $\square$

4.3. End-to-End Finite-Sample Bound

Theorem 6 (End-to-end sensitivity bound). *Suppose the conditions of Proposition 2, Theorem 4, and Theorem 5 hold. Then, jointly over the three independent samples, with probability at least $1 - \delta_D - \delta_E$,*

$$|\hat{\tau}_j^{\text{AIPW}} - \tau| \leq \varepsilon_{\text{AIPW},j} + \bar{\Gamma}_j \left[\hat{D}_j(\hat{\phi}_j) + r_{j,n_D}(\delta_D) \right]. \quad (89)$$

It also satisfies the oracle form

$$\begin{aligned} |\hat{\tau}_j^{\text{AIPW}} - \tau| &\leq \varepsilon_{\text{AIPW},j} \\ &\quad + \bar{\Gamma}_j \left[\inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j} \right]. \end{aligned} \quad (90)$$

Proof. The triangle inequality gives

$$|\hat{\tau}_j^{\text{AIPW}} - \tau| \leq |\hat{\tau}_j^{\text{AIPW}} - \tau_{\hat{Z}_j}| + |\tau_{\hat{Z}_j} - \tau|. \quad (91)$$

Theorem 5 bounds the first term. Theorem 3 and uniform candidate stability bound the second term by $\bar{\Gamma}_j D_j(\hat{\phi}_j)$. Proposition 2 gives

$$D_j(\hat{\phi}_j) \leq \hat{D}_j(\hat{\phi}_j) + r_{j,n_D}(\delta_D), \quad (92)$$

which proves equation (89). Equation (69) proves the oracle form. A union bound over the discrepancy-learning and evaluation events gives the stated joint probability. $\square$

Corollary 1 (Consistency). *Consider a sequence of candidate libraries, sample splits, nuisance estimators, optimization tolerances, and stability budgets satisfying Theorem 6. If*

$$\delta_{D,n} + \delta_{E,n} \rightarrow 0, \quad (93)$$

$$\frac{\log(2/\delta_{E,n})}{n_E} \rightarrow 0, \quad (94)$$

$$q_{e,j,n}(q_{0,j,n} + q_{1,j,n}) \xrightarrow{p} 0, \quad (95)$$

and

$$\bar{\Gamma}_{j,n} \left[\inf_{\phi \in \mathcal{F}_{j,n}} D_j(\phi) + 2r_{j,n_D}(\delta_{D,n}) + \varepsilon_{\text{opt},j,n} \right] \xrightarrow{p} 0, \quad (96)$$

then

$$\hat{\tau}_{j,n}^{\text{AIPW}} \xrightarrow{p} \tau. \quad (97)$$

Proof. The first three conditions make the evaluation-sample and nuisance-product components of

$\varepsilon_{\text{AIPW},j}$ converge to zero. Equation (96) makes the stability-weighted representation term vanish. The failure probability of Theorem 6 also converges to zero, which proves convergence in probability. $\square$

Remark 8 (Interpretation and observability). *Theorem 6 separates the final error into an AIPW estimation term and a hidden-confounding sensitivity term. The stability budget $\bar{\Gamma}_j$ and the nuisance errors $q_{e,j}$, $q_{0,j}$, and $q_{1,j}$ are not generally observed. The result becomes an operational numerical certificate after these quantities are bounded by additional theory, external information, or a declared sensitivity analysis.*

Remark 9 (Scope). *The theorem uses one prespecified held-out block, hard finite strata, a finite candidate library, and an honest three-way split. It does not establish a neural-class rate, a soft-to-hard approximation bound, an Adam optimization guarantee, or a cross-fitting theorem. Experimental aggregation across several held-out orientations is also outside this fixed- j result.*

5. Experiments

5.1. Questions, Comparators, and Reporting Rules

The experiments answer three separate questions. First, does proxy-supervised representation adjustment reduce ATE error relative to adjustment using X alone? Second, do the exact-assumption designs recover the known balancing structure? Third, does the method show a useful descriptive signal in an observational benchmark with a noisy randomized reference? We report mean absolute error, root mean squared error, paired wins, held-out discrepancy, overlap diagnostics, and Monte Carlo uncertainty. Oracle procedures that observe U or the true balancing score are simulation references, not implementable competitors. The theory in Sections 3 and 4 applies to one prespecified held-out block at a time. Experimental aggregation across several orientations is reported as an empirical extension and is not presented as a separate identification theorem.

5.2. Honest Approximate SCMs

The primary approximate study fixes the causal order $U \rightarrow (X, W)$, $(X, U) \rightarrow A$, and $(A, X, U) \rightarrow Y$. Both SCMs use $\dim(X) = 20$ and $\dim(W) = 10$, with twenty independently generated data sets and nested prefixes $n \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$. SCM1 uses linear proxy measurements and a nonlinear outcome component; SCM2 changes the structural equations to nonlinear transformations while preserving the same causal order and an informative monotone proxy channel. The proposed cross-mask method learns a representation from X and all but one proxy coordinate, audits balance with the held-out coordinate, and averages over masks.

At $n = 1000$, the mean absolute errors of proposed/ X -adjustment/full- (X, W) /oracle- (X, U) were 0.169/0.601/0.310/0.130 in SCM1 and 0.112/0.440/0.188/0.107 in SCM2. At $n = 10000$, the corresponding proposed errors were 0.199 and 0.134, versus 0.588 and 0.413 for X adjustment. These are finite-sample simulation results, not a proof of the audit-view identification theorem.

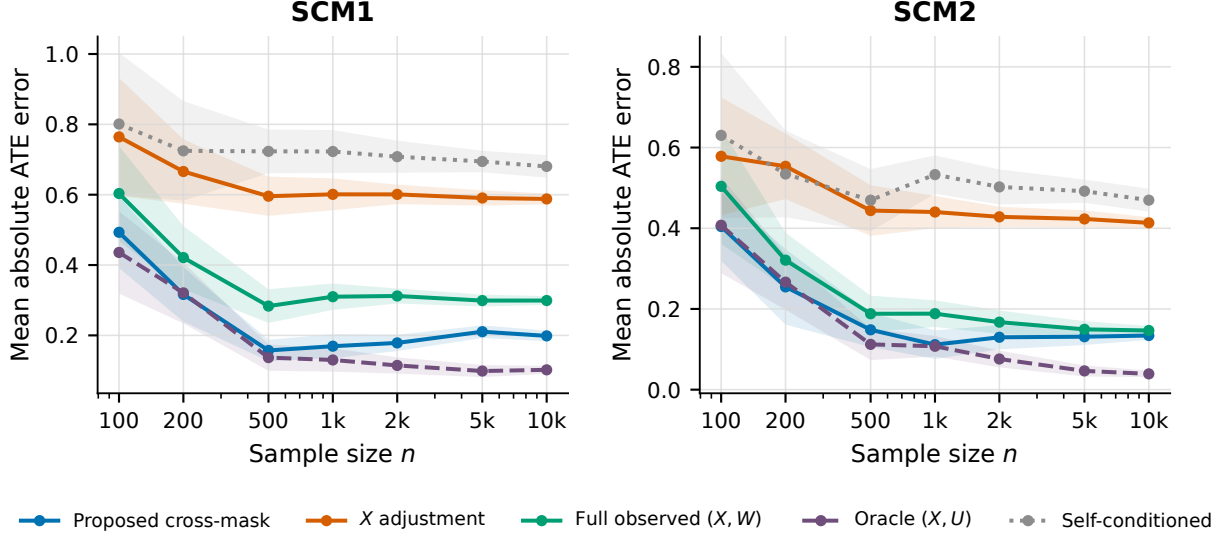


Figure 2: Honest approximate SCMs. Curves show mean absolute ATE error across twenty independently generated data sets; bands are approximate 95% Monte Carlo intervals for the mean. Prefixes are nested within a data set, so uncertainty is computed separately at each n and is not pooled across sample sizes. The proposed cross-mask learner uses one proxy coordinate for held-out auditing and averages across masks.

The self-conditioned heuristic remained substantially biased, which is consistent with the self-conditioning warning in Section 3.

5.3. Exact-Assumption Tracks A and B

Track A uses a finite balancing score, $\dim(X) = 20$, nine representation-side proxy coordinates, and one independent audit coordinate. The DGP enforces strict population overlap, exact decoding of the discrete confounding state from the representation-side view, and an injective Gaussian audit channel. At $n = 10000$ over twenty replications, mean absolute error was 0.175 for the learned audit representation, 0.433 for X adjustment, 0.186 for direct full-observed adjustment, and 0.033 for the oracle balancing score. The primary comparison with X passed, while the prespecified secondary superiority criterion against full-observed adjustment failed.

Track B uses a continuous exact balancing score and a linear representation class matched to that score. At $n = 10000$, mean absolute error was 0.0748 for the learned representation, 0.1583 for fitted X adjustment, and 0.1342 for fitted full-observed adjustment. The learned method beat X in 20/20 and full-observed adjustment in 19/20 paired replications. Its mean rank-recovery Spearman correlation was 0.915. The held-out conditional-dependence diagnostic used finite random features and is a surrogate, not an exact population KCI test; the design also covers one prespecified scalar- U SCM family.

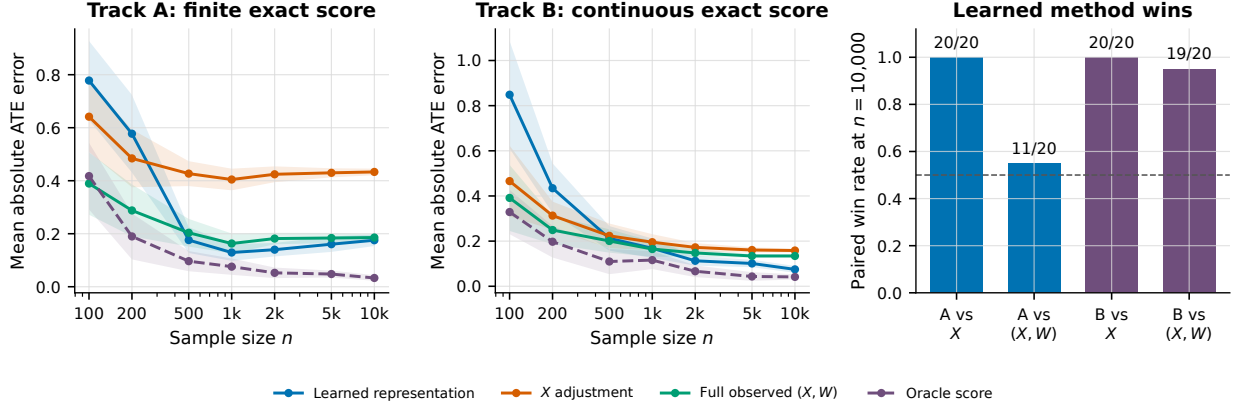


Figure 3: Exact-assumption experiments. The first two panels report mean absolute ATE error with approximate 95% Monte Carlo intervals over twenty replications. The third panel reports paired win rates at $n = 10,000$, where a win means that the learned representation has smaller absolute error than the indicated comparator. Track A satisfies the primary comparison with X but not the prespecified secondary superiority criterion against full observed adjustment. Track B records rank-recovery Spearman correlation 0.915 and held-out finite-RFF conditional-dependence surrogate 0.00691.

5.4. WSC Observational Benchmark

The WSC study forms an observational subset of 1,105 participants whose randomized mathematics assignment agrees with pretreatment preference, and targets the population ATE in the original 2,200 randomized participants. The full randomized difference in means is 0.760 with standard error 0.151 and serves only as a noisy benchmark. Across ten algorithmic split repetitions on this one fixed data set, the unconstrained cross-mask estimate averaged 1.302 with split standard deviation 0.065 and absolute benchmark gap 0.541, compared with gaps 2.690 for the naive observational contrast, 1.704 for direct X adjustment, and 0.672 for direct (X, W) adjustment. The unconstrained cross-mask method is the proposed WSC method shown in Figure 4. Its practical-overlap gate passed 0/10 repetitions. The benchmark therefore supplies descriptive evidence only, not a validated causal-performance claim, causal ground truth, or a test of proxy validity.

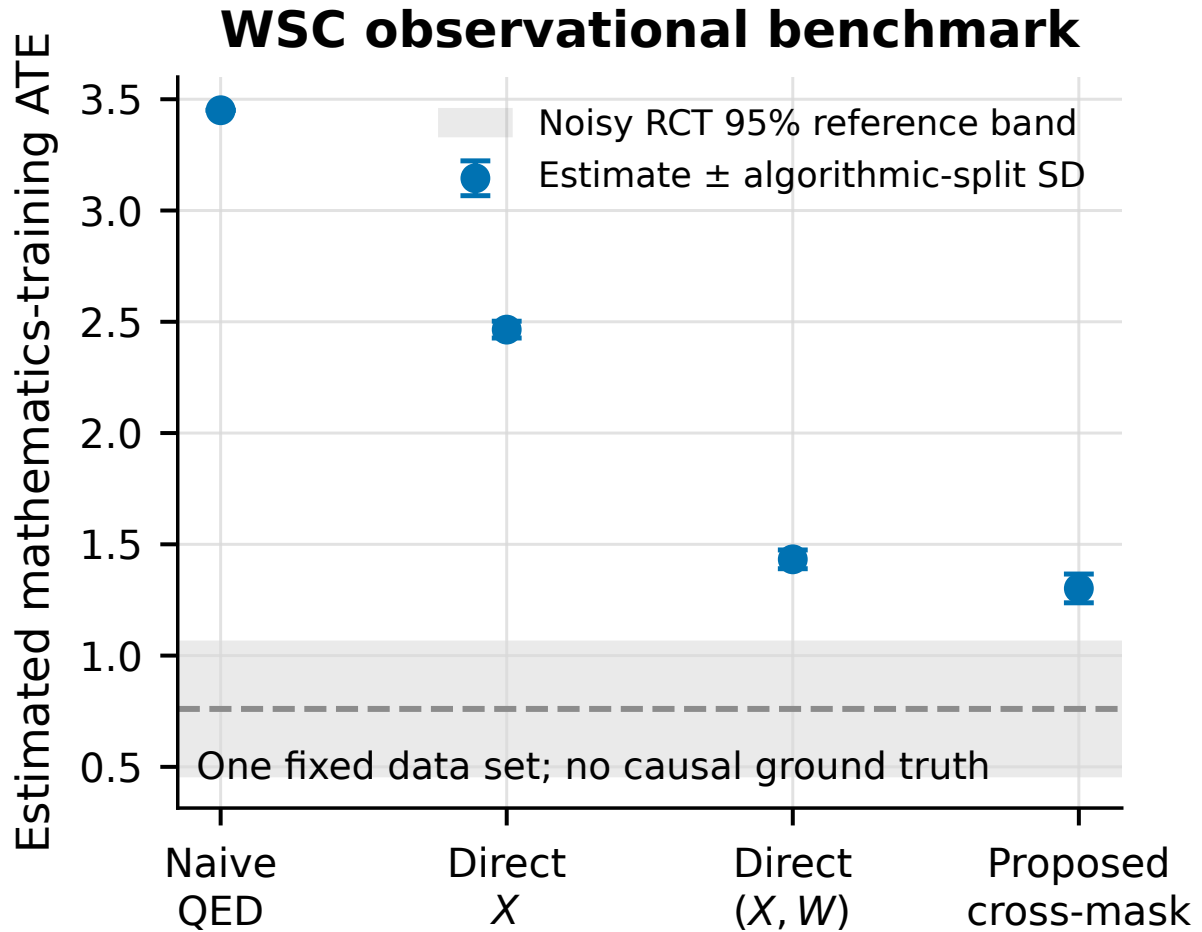


Figure 4: WSC observational benchmark. Estimates are shown relative to the noisy randomized reference. Error bars for fitted methods are standard deviations across ten algorithmic splits of one fixed data set, not Monte Carlo uncertainty. The benchmark has no causal ground truth and reports only the unconstrained cross-mask learner as the proposed method.

6. Discussion and Limitations

6.1. What the Method Delivers

For one prespecified block j , the canonical method uses W_{-j} to construct an outcome-free representation and uses the held-out block W_j to measure the residual imbalance of (X, W_j) across treatment arms within that representation. Under uniform outcome-relevant stability, the product $\Gamma_j D_j(\phi_j)$ is a sensitivity certificate for hidden-confounding bias. This is stronger than reporting a conditional-independence test alone because it targets a causal error scale, but weaker than point identification when $D_j(\phi_j) > 0$ because Γ_j is not generally identified from the observed distribution.

6.2. Testable and Structural Components

Held-out joint balance, empirical overlap, and minimum stratum mass are observable properties. Joint latent exchangeability, separation of the held-out proxy block, outcome-relevant completeness, and outcome-relevant stability are structural assumptions about the full-data law. Exact balance is therefore an observable population condition that can only be assessed up to sampling error, while its causal interpretation still requires structural assumptions. A large held-out discrepancy can falsify exact balance; a small test statistic cannot prove exact balance, completeness, or stability.

6.3. Attainability and Realized Bias

The exact identification theorem is conditional on the existence of a finite representation $Z_j = \phi_j(X, W_{-j})$ that attains exact cross-view balance. The theorem does not guarantee such attainability in a restricted representation class. In the approximate regime, reducing $D_j(\phi_j)$ reduces the certified worst-case bias under a common stability budget, but it need not reduce the realized bias when candidate representations have different stability constants or when the structural stability condition fails.

6.4. Learning and Inference Gaps

The current finite-sample theorem covers enumerable representation classes and fixed-fold evaluation. The neural experiments use soft assignments and covariance-weighted surrogate objectives that are not algebraically identical to the canonical MMD aggregation. A neural-class generalization bound, a soft-to-hard guarantee, and valid joint inference after adaptive representation selection remain open. These gaps prevent the manuscript from describing every neural result as a direct verification of the theory, even when its empirical error is lower.

6.5. Audit Views and Practical Scope

Cross-view learning uses one prespecified proxy block for held-out evaluation and the remaining high-dimensional measurement for representation learning. This design blocks literal self-copying of the evaluated block. It does not by itself establish separation among proxy views, so point identification still needs the structural assumptions in Section 3. The experiments repeat the procedure

across several held-out orientations and aggregate their estimates as an empirical extension; the fixed- j theory applies separately to each orientation. The WSC experiment further shows that an encouraging benchmark gap can coexist with a failed practical-overlap rule. Real-data evidence must therefore report overlap, held-out discrepancies, and benchmark uncertainty without treating any one diagnostic as ground truth.

6.6. Scope of the Contribution

The defensible contribution is the combination of a cross-view proxy-supervised representation, an observable conditional discrepancy, an exact identification theorem, a stability-indexed approximate guarantee, and an honest finite-sample estimation analysis. The paper does not claim that one generic correlated proxy point-identifies the ATE, that every smaller discrepancy has smaller realized bias without a common stability budget, or that the current neural optimizer reaches the population optimum. A systematic literature audit and broader external validation are required before any firstness or universality claim.

7. Conclusion

Single proxy balancing asks what a structured high-dimensional pretreatment proxy can reveal about hidden-confounding bias after learning a low-dimensional adjustment representation. For one prespecified held-out block j , the cross-view construction learns $Z_j = \phi_j(X, W_{-j})$ and evaluates treatment balance through (X, W_j) . Exact cross-view balance identifies the ATE under the stated structural assumptions, while approximate balance yields the sensitivity certificate $|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j)$. Empirical discrepancy minimization and honest AIPW estimation give an end-to-end bound that separates learning, hidden-confounding, nuisance, and evaluation errors. The current simulations support the method in exact and approximate designs while exposing limitations from unattainable balance, weak proxy channels, surrogate mismatch, and practical-overlap loss. Neural-class generalization, soft-to-hard transfer, stability calibration, and external validation remain necessary for a submission-ready paper.

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A. Simulation Details

The existing project-local experiments will be documented by recording each data-generating process, parameter value, sample size, replication count, and random seed policy. Every manuscript figure and table will be linked to the script and machine-readable result from which it was produced.

The simulation audit will separate observed quantities from oracle quantities. In particular, any result that uses U or a direction constructed from U will be labeled as an oracle diagnostic and will not be presented as the output of a learnable procedure.

B. Proof Obligations

The first proof obligation is the conditioning-safety argument for $Z_j = \phi_j(X, W_{-j})$. It must show how joint latent exchangeability and separation of the held-out block remain valid after conditioning on the learned representation.

The second proof obligation is the exact bias decomposition and cross-view factorization. These results must connect the observed adjustment functional, the two treatment-arm latent laws, and the held-out distribution through one common proxy channel.

The third proof obligation is exact identification under outcome-relevant completeness. It must show that exact observed balance makes the two outcome-relevant latent averages equal without asserting equality of the full latent distributions.

The fourth proof obligation is the approximate sensitivity bound. It must show how the stability constant converts each held-out discrepancy into an outcome-relevant latent bound and how the stratum weights aggregate to $\Gamma_j D_j(\phi_j)$.

The fifth proof obligation is the finite-sample chain from uniform discrepancy concentration through empirical representation selection to honest AIPW estimation. Its final inequality must display the representation, stability, nuisance, and evaluation-sample terms separately.